HW4: PM2.5 in Northern Thailand

From Raw Data to a Next-Day Exceedance Warning Model

Study area: Chiang Mai and Chiang Rai | **Period:** 2023–2025
Target: Whether tomorrow's daily mean PM2.5 exceeds 37.5 µg/m³

Executive Summary

This report investigates whether today's PM2.5 and same-day weather conditions can help predict whether tomorrow's daily mean PM2.5 will exceed Thailand's 24-hour ambient standard of 37.5 µg/m³. The analysis uses two Open-Meteo endpoints—Air Quality and Historical Weather—and joins them by local calendar date and location. Air4Thai is used as an independent ground-station source for the required spot-check, subject to its current-only endpoint limitation.

The final modelling table contains **2,176 daily observations**: 1,088 from each location. Overall mean PM2.5 is **19.98 µg/m³**, the maximum is **109.92 µg/m³**, and only **182 observations (8.36%)** are positive exceedance cases. PM2.5 is strongly seasonal, with the most pronounced peaks in March. Chiang Mai has the higher mean concentration (21.57 µg/m³), while Chiang Rai has the higher exceedance rate (9.38% versus 7.35%).

A class-weighted logistic regression is evaluated using a chronological 80/20 hold-out and 5-fold TimeSeriesSplit. The final test period contains no positive cases, making its 99.77% accuracy unsuitable as evidence of exceedance detection. Across ordered validation folds, mean recall is **0.582** and mean F1 is **0.411**. The recommendation therefore treats the model as an additional early-warning signal rather than an automatic decision maker.

Key message: The data support a seasonal, location-aware monitoring strategy, but the present model is not reliable enough to be the sole basis for public warnings.

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1. Problem Background

Fine particulate matter (PM2.5) consists of particles with an aerodynamic diameter of 2.5 micrometres or smaller. Because these particles can penetrate deeply into the lungs and can enter the bloodstream, exposure is associated with cardiovascular and respiratory harms. The World Health Organization identifies particulate matter as one of the air pollutants with the strongest evidence for adverse health outcomes. [1]

Northern Thailand experiences recurring periods of elevated particulate pollution, particularly during the dry season. The course brief describes a combination of agricultural residue burning, forest fires, transboundary smoke, stable atmospheric conditions, and mountain topography that can trap pollution in valleys. These conditions make timely information useful for local public-health and emergency-management decisions.

For this project, the operational threshold is Thailand's 24-hour ambient PM2.5 standard of **37.5 μg/m³**. The Pollution Control Department states that the 24-hour standard changed from 50 to 37.5 μg/m³ from 1 June 2023. [2] This threshold is used here as the classification boundary, not as a claim that concentrations below it are risk-free. WHO's 2021 health-based guideline is substantially lower, illustrating that regulatory standards and health guidelines serve different purposes. [1]

Who is affected?

High PM2.5 can affect the general population, but exposure is especially important for children, older adults, pregnant people, people with existing cardiopulmonary conditions, outdoor workers, and others who spend substantial time outdoors. The practical decision problem is therefore not simply whether pollution exists, but whether authorities can identify higher-risk days early enough to support protective communication and preparedness.

Why this analysis matters

A statement such as “PM2.5 is bad in March” is already widely known. The analytical contribution here is narrower and testable: quantify when exceedances occur, determine whether the two selected locations behave similarly, identify weather conditions associated with next-day exceedances, and test whether information available at the end of today can improve a binary warning decision for tomorrow.

Sources: [1] World Health Organization, Global Air Quality Guidelines (2021). [2] Pollution Control Department / Environmental and Pollution Control Office 3, announcement on Thailand's PM2.5 standard.

2. Approach

Research question

Can today's PM2.5 level and same-day weather conditions predict whether tomorrow's daily mean PM2.5 will exceed 37.5 µg/m³ in Chiang Mai and Chiang Rai?

Why this question?

The question is designed around a decision that could be acted on one day ahead. A numeric forecast would be useful for planning, but a classification target maps directly to a warning/no-warning decision. Recent PM2.5 values capture pollution persistence, while weather variables provide information about atmospheric conditions that may accompany high-pollution days.

We selected **Chiang Mai and Chiang Rai** rather than Chiang Mai alone to create a controlled spatial comparison. Both locations have the same requested date range and the same set of variables, allowing us to ask whether the problem looks the same in two Northern Thailand locations and whether location can be used as a predictive feature.

What we expected before analysis

Before inspecting the final results, we expected a strong seasonal pattern with higher PM2.5 during the dry-season months. We also expected recent PM2.5 levels and weather conditions such as precipitation, humidity, wind, and temperature to contain some predictive information about the following day's exceedance status. We did not assume that any single weather variable would be causal.

Analytical framing

Framing	Classification
Target	Whether tomorrow's daily mean PM2.5 exceeds 37.5 µg/m³
Why it matters	Direct input to an early-warning decision
Prediction time	End of today's daily observation
Locations	Chiang Mai and Chiang Rai

A false negative is a missed high-PM2.5 day, which is more consequential for a warning application than a small error in a numeric concentration forecast.

3. Method

3.1 Data sources and acquisition

Data were fetched programmatically and raw responses were saved before preprocessing. The main historical dataset uses two Open-Meteo endpoints with the same coordinates, date range, and Asia/Bangkok timezone. Air4Thai provides current ground-station observations and is kept separate from the historical modelling join.

Source	Endpoint / role	Main variables	Coverage
Open-Meteo Air Quality	/v1/air-quality	PM2.5, PM10, CO, dust	2023-01-01–2025-12-31; hourly
Open-Meteo Historical Weather	/v1/archive	Temperature, RH, wind, precipitation, pressure	2023-01-01–2025-12-31; hourly
Air4Thai / PCD	getNewAQI_JSON.php	Current station PM2.5	Current snapshot; no historical endpoint

Fetch and raw-data audit

Location	Air hourly	Weather hourly	Air daily	Weather daily	Unmatched daily dates
Chiang Mai	26,304	26,304	1,096	1,096	0
Chiang Rai	26,304	26,304	1,096	1,096	0

The acquisition script is re-runnable. Historical requests use a fixed date range and should be reproducible for a given provider version, although historical model/reanalysis products can be revised. Air4Thai is different: its endpoint returns the current reading, so running the script on different dates can legitimately produce different current files. The raw response is retained so the retrieval snapshot can be audited.

3.2 Time verification (C1)

Both historical endpoints were requested with **timezone=Asia/Bangkok**. We verified agreement by comparing their hourly axes, checking the first and last timestamps, and comparing the resulting local-date sets after daily aggregation. Both endpoints returned 26,304 hourly rows per location, from 2023-01-01 00:00 through 2025-12-31 23:00, and produced 1,096 daily dates per location with zero unmatched dates. Thus the daily averages are based on Thailand local calendar days and the two endpoint time axes agree before joining.

3. Method (continued)

3.3 Cleaning, joining, and feature construction

Hourly observations from the two Open-Meteo endpoints were aligned by local timestamp and aggregated to daily values. The resulting daily tables were joined by **date + location**. The joined daily dataset contained 1,096 rows for each location and zero unmatched dates. Lagged variables and the next-day target were then constructed. After removing rows that cannot have the required history or next-day outcome, the final modelling table contains 2,176 rows: 1,088 per location.

Stage	Result
Hourly air-quality data	26,304 rows/location
Hourly weather data	26,304 rows/location
Daily aggregation	1,096 rows/location
Daily join	1,096 rows/location; 0 unmatched dates
After lag/rolling + next-day target construction	1,088 rows/location
Final model table	2,176 rows; 20 columns

3.4 Features and target

Feature group	Variables	Rationale
Recent pollution	pm25_today, pm25_lag1, pm25_lag7, pm25_7d_mean	Persistence and recent pollution history
Air quality	pm10, dust	Related particulate/aerosol information
Weather	temperature, relative humidity, wind speed, precipitation, surface pressure	Same-day atmospheric context
Context	location, day_of_week, month	Spatial and temporal structure

The target **tomorrow_exceeds_37_5** equals 1 when tomorrow's daily mean PM2.5 is greater than 37.5 µg/m³ and 0 otherwise. The final table has 1,994 negatives (91.64%) and 182 positives (8.36%), so class imbalance is substantial.

3.5 Split, preprocessing, models, and baseline

The data were sorted chronologically before splitting. The first 80% (1,740 rows; 2023-01-08 to 2025-05-26) formed the training period and the final 20% (436 rows; 2025-05-27 to 2025-12-30) formed the hold-out test period. Within the training period, 5-fold TimeSeriesSplit was used so validation always occurs after the corresponding training data. Numerical features were median-imputed and standardized; location was one-hot encoded. The model is logistic regression with balanced class weights. The baseline is the majority-class classifier.

4. Results

4.1 When is the problem worst?

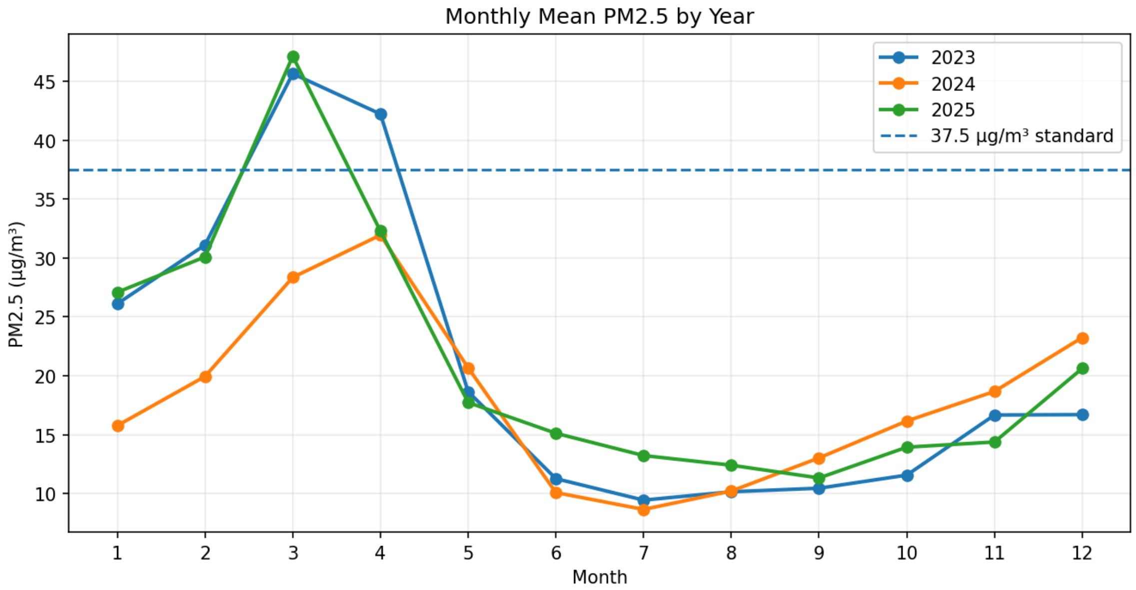

Figure 1. Monthly mean PM2.5 by year. *The x-axis is month and the y-axis is mean PM2.5 (µg/m³); the dashed line marks 37.5 µg/m³. The figure shows a strong seasonal pattern. March is the clearest peak: 45.63 µg/m³ in 2023 and 47.11 µg/m³ in 2025, both above the standard, while concentrations are much lower through much of the middle of the year. The magnitude of the peak varies by year, so month is informative but not sufficient by itself.*

4.2 How many days exceed the standard?

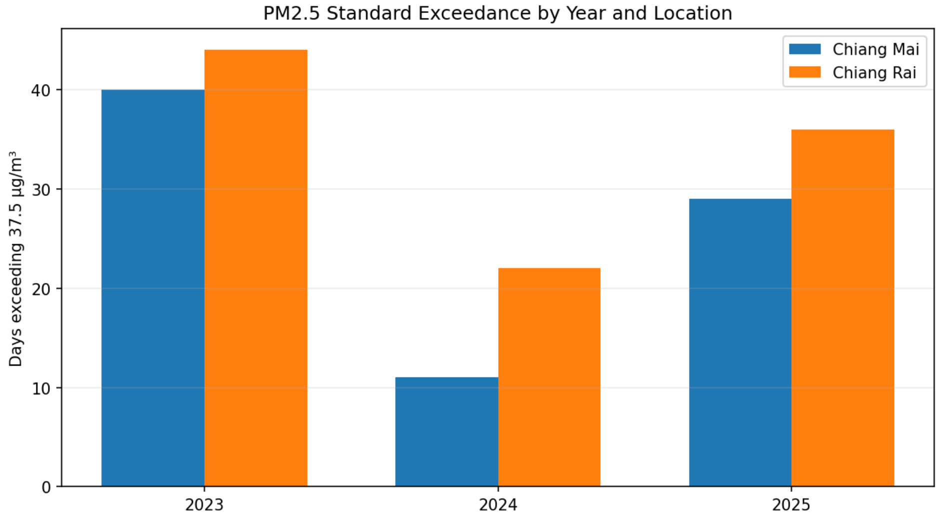

Figure 2. Number of days with daily mean PM2.5 above 37.5 µg/m³ by year and location. *Exceedance days were 40/44 (Chiang Mai/Chiang Rai) in 2023, 11/22 in 2024, and 29/36 in 2025. The pattern is not a monotonic upward or downward trend: exceedances fell sharply in 2024 and increased again in 2025. Chiang Rai had more exceedance days in each year shown.*

4. Results (continued)

4.3 Does the problem look the same in different places?

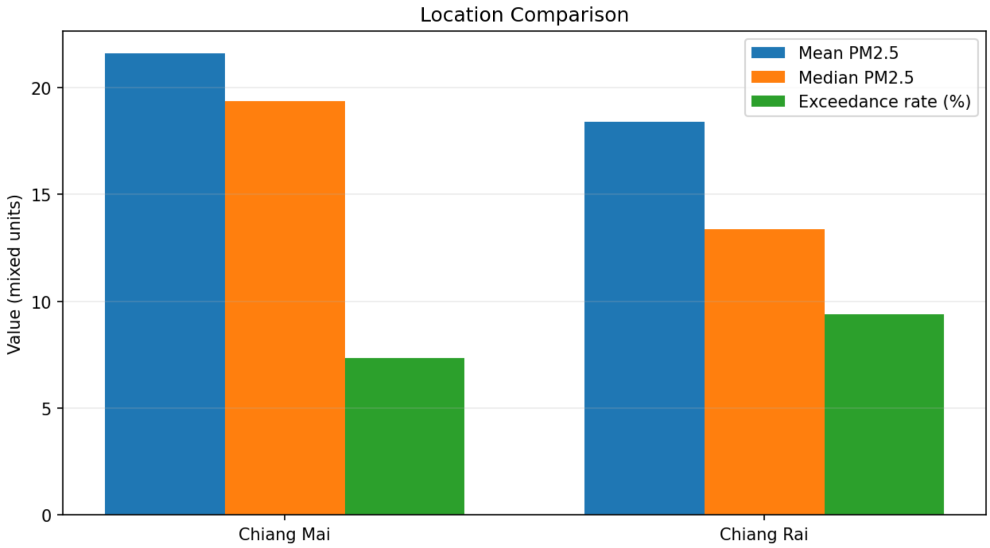

Figure 3. Location comparison. *Chiang Mai has a higher mean PM2.5 (21.57 µg/m³) and median (19.36 µg/m³) than Chiang Rai (18.39 and 13.35 µg/m³). However, Chiang Rai has the higher exceedance rate, 9.38% versus 7.35%. Thus "higher typical concentration" and "more frequent standard exceedance" are not identical conclusions. This spatial difference supports keeping location in the model.*

4.4 Which weather conditions accompany the worst days?

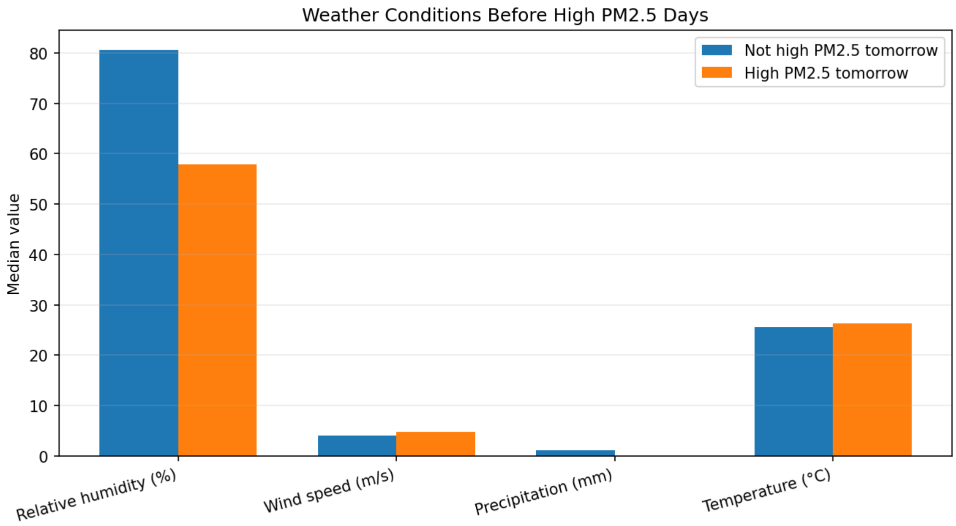

Figure 4. Median weather conditions on days before high-PM2.5-tomorrow versus non-high days. *High-PM2.5-tomorrow observations have lower median relative humidity (57.94% versus 80.54%) and zero median precipitation. Median wind speed is slightly higher (4.69 versus 4.08 m/s), and median temperature is slightly higher (26.34 versus 25.51°C). These are associations, not evidence that any one weather variable causes the exceedance.*

4. Results (continued)

4.5 Is there a weekly pattern?

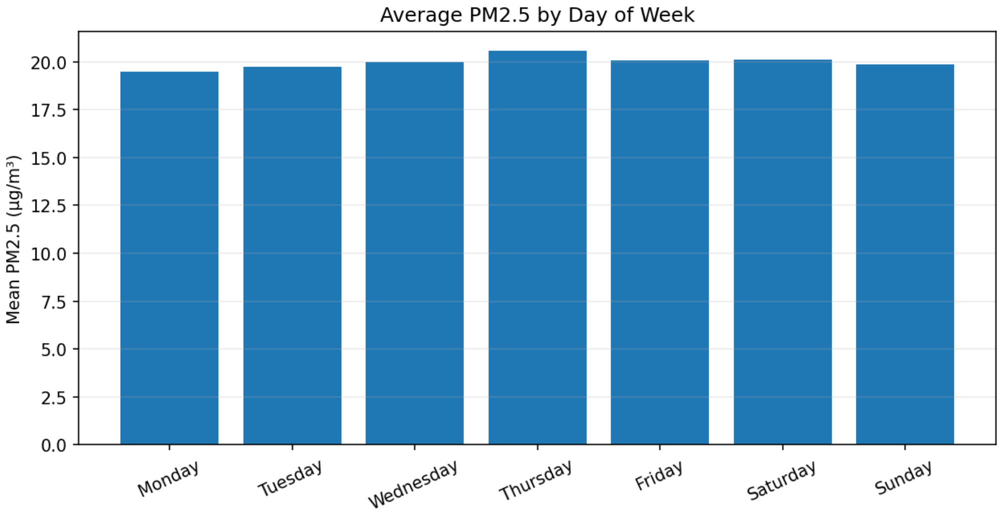

Figure 5. Average daily PM2.5 by day of week. *Thursday is highest at 20.57 µg/m³ and Monday is lowest at 19.50 µg/m³. The difference is small compared with the seasonal variation in Figure 1, so the weekly pattern is weak. Day of week is retained as a secondary model feature rather than treated as a major driver.*

4.6 Model performance versus baseline (C5)

Metric	Majority baseline	Logistic regression
Accuracy	1.0000*	0.9977*
Balanced accuracy	1.0000*	0.9977*
Positive precision	0.0000*	0.0000*
Positive recall	0.0000*	0.0000*
Positive F1	0.0000*	0.0000*
ROC-AUC	N/A*	N/A*

* The final test period contains 436 observations and **zero positive cases**. Therefore the apparent 99.77% test accuracy does not demonstrate useful exceedance detection.

Ordered cross-validation: Mean accuracy 0.9131, balanced accuracy 0.8450, precision 0.3482, recall 0.5821, and F1 0.4114 across five TimeSeriesSplit folds. These results are the more informative evidence for positive-class performance.

4.7 Six Checkpoint Answers

C1 · Time

Both Open-Meteo endpoints were requested in Asia/Bangkok time. We verified agreement by comparing their hourly axes (26,304 observations/location, 2023-01-01 00:00 through 2025-12-31 23:00) and their daily date sets after aggregation. Each produced 1,096 daily dates/location and the join had zero unmatched dates. The daily averages therefore correspond to Thailand local calendar days.

C2 · The join

Before joining, each location had 1,096 daily air-quality rows and 1,096 daily weather rows (26,304 hourly rows in each source). The join produced 1,096 rows/location with zero unmatched dates. The final 1,088 rows/location are explained by feature construction: the first seven days cannot supply the full lag/rolling history and the final day cannot supply a next-day target.

C3 · Missing values

After construction of the final modelling table, all 20 model columns had 0.00% missing values. This does not mean the raw APIs are intrinsically complete; it means preprocessing produced a complete modelling table after alignment, feature construction, and removal of rows without the required history/next-day information.

C4 · Comparing places

The locations return materially different values. Chiang Mai has mean/median PM2.5 of 21.57/19.36 µg/m³, while Chiang Rai has 18.39/13.35 µg/m³. Exceedance rates are 7.35% and 9.38%, respectively. These differences are evidence that the two locations do not produce identical PM2.5 behaviour.

C5 · Model versus baseline

On the same final test set, the majority baseline has 1.0000 accuracy and logistic regression has 0.9977. However, the test set has no positive observations, so positive-class metrics are not informative. TimeSeriesSplit gives mean recall 0.5821 and F1 0.4114, which is the appropriate evidence for positive-event detection.

C6 · Ground truth

Air4Thai provides an independent ground-station spot-check. The closest available stations to the project coordinates were used: station 36t for Chiang Mai and station 57t for Chiang Rai. Both readings are timestamped 2026-08-31 at 15:00.

C6 comparison: Air4Thai value → project value → difference

Location	Air4Thai value	Project value	Difference
Chiang Mai	10.3 µg/m³ (36t; 15:00)	21.57 µg/m³ (2023–2025 daily mean)	–11.27 µg/m³
Chiang Rai	7.3 µg/m³ (57t; 15:00)	18.39 µg/m³ (2023–2025 daily mean)	–11.09 µg/m³

Difference calculation

Difference = Air4Thai value – project historical location mean. Chiang Mai: 10.3 – 21.57 = –11.27 µg/m³. Chiang Rai: 7.3 – 18.39 = –11.09 µg/m³.

Why they disagree

The Air4Thai values are single-station hourly observations on 2026-08-31, whereas the project values are historical daily means over 2023–2025 at Open-Meteo grid points. Therefore this is an independent ground-truth sanity check, not a same-time model validation. The disagreement can arise from different dates/aggregation windows, station-versus-grid representation, and spatial mismatch between a ground monitor and a gridded model point.

5. Conclusion

5.1 What we found

The analysis shows a strong seasonal PM2.5 pattern in Chiang Mai and Chiang Rai, with the most pronounced concentrations around March. The annual number of exceedance days varies substantially from year to year rather than following a simple trend. The two locations are not interchangeable: Chiang Mai has the higher typical PM2.5 level, while Chiang Rai has the higher proportion of days above 37.5 µg/m³. Weather conditions also differ between days preceding high and non-high next-day PM2.5, especially for humidity and precipitation.

The machine-learning result is deliberately reported with caution. The final chronological test period contains no positive events, so its near-perfect accuracy is not evidence that the classifier can detect dangerous days. Ordered cross-validation gives moderate positive-event detection, with mean recall of 58.2% and F1 of 41.1%. In a warning context, false negatives remain a serious concern.

5.2 Recommendation and audience

Audience: provincial public-health and disaster-management authorities in Chiang Mai and Chiang Rai.

Use a **seasonal, location-specific early-warning workflow** during the higher-risk dry-season period. Today's PM2.5, recent PM2.5 history, and same-day weather conditions can be used as an additional screening signal for tomorrow's risk. When the model flags elevated risk, authorities can prioritize monitoring, public communication, and preparedness rather than waiting for the next day's observed concentration.

5.3 What the analysis does not support

The evidence does not establish that low humidity or lack of rain causes high PM2.5; the analysis is observational. It also does not justify replacing official monitoring with the model, because the model misses some positive events and the hold-out test set is not informative for positive detection. Results from two locations should not be generalized to all Northern Thailand without additional station-level validation.

5.4 Limitations and next steps

The main limitations are class imbalance, the absence of positive cases in the final hold-out period, only two locations, modelled/reanalysis inputs for the two Open-Meteo historical sources, and the current-only nature of Air4Thai. With more time, the strongest next step would be to collect a historical ground-station dataset across more Northern Thailand stations and construct a future test window containing both positive and negative cases while preserving temporal order.

Bottom line: The data earned a recommendation for targeted early warning and monitoring, not a claim that the current classifier is ready for autonomous public alerts.

Appendix A1 · Data Quality and Reproducibility

Final modelling columns and missingness

Column	Missing
date	0.00%
pm2_5	0.00%
pm10	0.00%
carbon_monoxide	0.00%
dust	0.00%
temperature_2m	0.00%
relative_humidity_2m	0.00%
wind_speed_10m	0.00%
wind_direction_10m	0.00%
precipitation	0.00%
surface_pressure	0.00%
location	0.00%
pm25_today	0.00%
pm25_lag1	0.00%
pm25_lag7	0.00%
pm25_7d_mean	0.00%
pm25_tomorrow	0.00%
tomorrow_exceeds_37_5	0.00%
day_of_week	0.00%
month	0.00%

Reproduction workflow

Run the repository in order: (1) fetch raw data, (2) prepare/clean/join data, (3) generate descriptive figures, and (4) train/evaluate the model. Raw JSON files are stored in data/raw/; the processed daily table is stored in data/processed/; figures are stored in outputs/figures/; and model metrics are stored in outputs/results/. The report values are intended to be traceable to these outputs.

Key retrieval parameters

Parameter	Value
Chiang Mai	18.7883, 98.9853
Chiang Rai	19.9105, 99.8406
Timezone	Asia/Bangkok
Date range	2023-01-01 to 2025-12-31
PM2.5 threshold	37.5 µg/m³, 24-hour mean
Final model date range	2023-01-08 to 2025-12-30

References

[1] World Health Organization. Global Air Quality Guidelines: particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. 2021. https://www.who.int/publications/i/item/9789240034228

[2] Pollution Control Department / Environmental and Pollution Control Office 3. Announcement on the ambient PM2.5 standard. https://epo03.pcd.go.th/th/news/detail/154265

[3] Open-Meteo. Air Quality API documentation. https://open-meteo.com/en/docs/air-quality-api

[4] Open-Meteo. Historical Weather API documentation. https://open-meteo.com/en/docs/historical-weather-api

[5] Pollution Control Department / Air4Thai. Current station readings endpoint. http://air4thai.pcd.go.th/services/getNewAQI_JSON.php

Appendix A2 · AI Disclosure and Rubric Audit

AI-use disclosure

AI assistance was used for explanation, report organization, wording refinement, debugging guidance, and review of the analysis against the assignment rubric. Numerical results were taken from the project's analysis outputs. Unsupported claims were removed, and the unresolved C6 checkpoint is explicitly disclosed rather than filled with an invented value.

Where AI was used

Area	What was asked	What was changed / rejected
Report structure	Map project evidence to Sections 1–5 and C1–C6.	Reorganized into the fixed rubric structure; unsupported claims removed.
Method explanation	Explain time alignment, joins, chronological splitting, and imbalance.	Checked against observed row counts and model outputs.
Results interpretation	Explain figures and model metrics.	Kept interpretations tied to produced figures; avoided causal claims.
Recommendation	Turn findings into a concrete decision-maker recommendation.	Restricted to an additional early-warning signal; no autonomous-warning claim.
C6 ground truth	Identify evidence needed for Air4Thai comparison.	Added station values, project values, explicit differences, and the explanation of disagreement.

Rubric coverage audit

Requirement	Status
Fixed structure: Sections 1–5	Covered; 10 main pages excluding appendix
Part A: coded acquisition, raw preservation, metadata	Documented in Method; repository scripts/outputs referenced
Part B: at least 3 interpreted figures	5 figures, each captioned and interpreted
C1 Time	Answered with timezone + hourly/daily axis verification
C2 Join	Answered with before/after row counts and zero unmatched dates
C3 Missing values	All 20 final modelling columns reported at 0.00%
C4 Comparing places	Answered with numerical evidence
C5 Model vs baseline	Side-by-side metrics plus TimeSeriesSplit explanation
C6 Ground truth	Completed with Air4Thai → project → difference → disagreement explanation
Part D recommendation	Named audience + concrete use + false-negative consequence
Results that did not work out	Final test has no positives; C6 same-time evidence limitation disclosed
AI disclosure	Completed on this final appendix page

C6 evidence limitation

The available Air4Thai endpoint provides a current station snapshot, while the modelling period ends in 2025. Consequently, the supplied files do not support an exact same-date 2023–2025 Air4Thai-vs-project numerical validation. The report therefore labels the comparison as a ground-truth sanity check and does not fabricate a historical station value.